Critical Review of Literature on the Interaction of Social Sciences in Contemporary Marine Studies

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Introduction

The convergence of social sciences with other branches of human knowledge is a significant aspect of the history of academic disciplines. Rather than arbitrary intersections between two distinct and independent paths, we observe intertwined developments that connect the context of society and the individual with both technological materiality and the most elusive conceptual abstractions. For example, investigations into cellular development during a particular era provide metaphors for observations on social rise or decline (Esposito, 2014); conversely, the identification of numerical regularities in population records, for instance, laid the groundwork for the probabilistic understanding of natural phenomena (Hacking, 1990).

The field of ocean sciences, or oceanography, has amassed a vast body of knowledge since its foundation in the late 19th century. A snapshot of the accumulated knowledge about the sea in 1942 is encapsulated in the book *The Oceans. Their Physics, Chemistry and General Biology* (Sverdrup et al.), which contained 1,087 pages and seven attached maps. Today, the *Encyclopedia of Ocean Sciences* comprises six volumes with a total of 4,361 pages (Steele et al., 2001). This increase reflects a theoretical body encapsulated in an encyclopedic format, adding to the current volume of oceanographic data: the equivalent of a digital duplicate of the ocean itself[1] (Lehman, 2016).

Expanding the scope to account for the relationships between techno-scientific knowledge and the sociopolitical life that sustains it (particularly in relation to the United Nations' Sustainable Development Goals), we find a starting point in the mid-20th century. During this period, questions emerged such as: Is it possible to reorient the development of science and technology, which seems to be autonomous? Does control over nature also imply control over human beings? How does the scientific worldview impact traditional institutional frameworks? In essence, under what conditions will humans (divided in so many ways) be able to capable not only of avoiding destruction but of living together relatively well on the planet? (Habermas, 1968; Hoffman, 1966).

[1] Only the National Oceanic and Atmospheric Administration (NOAA) of the USA has a catalog with 91,948 data sets (https://data.noaa.gov/dataset/)

The incorporation of these questions into decision-making models aimed at linking knowledge with action to achieve interconnected sociopolitical goals, along with the pursuit of increasingly consensual knowledge, marks the origin of the relationship between science and technology for environmental management (Haas et al., 1977). This expansion and interconnection of sociopolitical objectives, crystallized in planning, have called upon the social sciences from the beginning[2], perhaps as disciplines that closely relate to symbolically constructed domains. Today, this call comes from contemporary marine studies.

While concerns about aquatic resources and spaces are not new to humanity, the extent, intensity, and diversity of these concerns today are unprecedented: food, hydrocarbons, minerals, desalinated water, ornaments, genetic resources, scientific information, navigation, cabling and pipelines, tourism and recreation, marine energies, geoengineering, waste management, conservation, territorial limits, and military activities (Jouffray et al., 2020; Kolodziej, 2020). Equally diverse are the perspectives from which social sciences are asked to intervene in this array of interests, from their strongholds in sociology, political science, law, psychology, anthropology, ethnography, history, and geography.

This critical literature review accounts for an academic discussion that has been grouped under the labels of Marine Social Sciences, Maritime Sociology, Sociology of the Oceans, and Social (or Human) Oceanography. It also considers writings that, without suggesting a "new discipline" in their titles, emphasize the expected contributions of social sciences to fisheries, coastal management, and, more broadly, the ecosystem approach. This review aims to summarize these proposals, suggesting ways to group them according to their interdisciplinary ambition, reasoning styles, or comprehensive view of the issue.

[2] Furthermore, the institutional incorporation of Social Sciences into the international order can be observed in the following sequence of departments within the United Nations Educational, Scientific and Cultural Organization (UNESCO): Social Sciences and Philosophy/Social Sciences Section 1946-1948; Department of Social Sciences, 1948-1965; Department of Social Sciences within the Social Sciences, Humanities and Culture Sector (SHC), 1965-1974; Sector for Social Sciences and their Applications, 1976-1984; and, Social and Human Sciences Sector (SHS), 1984-present day. (https://atom.archives.unesco.org/unesco-social-and-human-science-sector-and-its-predecessors).

Method

This compilation was conducted through a search in the databases of Science Direct, Sage Journals, Springer Link, Jstor, Taylor & Francis Group, and on the platforms of Academia and Research Gate. A total of 34 academic articles, 5 book chapters, and complementary documents[3] related to the theme of social sciences and marine studies were reviewed. The article that prompted this critical review was "Marine Social Sciences: Looking Towards a Sustainable Future" (2020) by Emma McKingley, Tim Acott, and Katherine L. Yates. From this, its bibliography was examined, and those works specifically mentioning a link between social sciences and marine studies were selected. For this citation tree exercise, the software Citation Tree (https://www.citationtree.org/), which draws on data from Crossref and Semantic Scholar, was used.

To organize the selected material, the articles were thematically categorized according to their focus and scope: the conceptualization issues of marine social sciences, their most representative historical episodes, the type of theoretical framework they fit into, and the experience of fisheries.

An Ocean of Perspectives

In 1908, the inaugural issue of the *Internationale Revue der gesamten Hydrobiologie und Hydrographie* noted that the spheres of biological and geological-hydrographic knowledge of waters were inseparable because "...whether in the form of rivers, lakes, or oceans, they are never a factor that shapes the earth without simultaneously being a medium for life, and conversely, they are never a medium for life without exerting a significant influence in shaping the earth's surface" (Sverdrup et al., 1942, p. 1). In a similar vein, 113 years later, the United Nations' program "Decade of Ocean Science for Sustainable Development 2021-2030" asserts that "The term 'ocean science' includes disciplines from both natural and social sciences, encompassing interdisciplinary topics; the technology and infrastructure that support marine sciences; the application of this science for societal benefit, including knowledge transfer and applications in regions lacking scientific capacity; and the interfaces of science with public policy and innovation […]. Ocean science recognizes, respects, and embraces local and indigenous knowledge" (Implementation Plan Summary, 2021, p. 11).

[3] Among them: 1 book project, 1 column, 4 conference programs, 2 conference calls for papers, 4 editorials, 1 manifesto, 1 memorandum, 2 book reviews, and 1 summary.

While this marks an institutional milestone in recognition, the concept has been addressed in various academic circles for over two decades, whether with an emphasis on defining a maritime-social space (Steinberg, 1999; Gordinier, 2015; Crockford, 2020; Lambach, 2022), from the perspective of sociology (Poole, 1981; Kolodziej et al., 2010; Cocco, 2013; Longo et al., 2016; Hannigan, 2017; Picken, n.d.), through the creation of research communities (MARE Conference; McKinley et al., 2022; Narchi et al., 2019), or through an interest in fisheries (Symes et al., 2010; Jentoft et al., 2014; Delgado et al., 2022). This notable intellectual productivity aligns with the role, qualities, and expectations placed on what we now call the global ocean: engine of development, origin and condition of life, thermal regulator, resource source, and final frontier. If each of these labels reflects the interests of our societies (culture, social institutions, and individuals)[4], it is not surprising that each dimension of knowledge has something to say about the sea.

Within this ocean of viewpoints lies an underlying orientation: the ambition to frame the relationship between society and nature differently (Longo, 2016). Although almost the entire vision is approached from the perspective of sustainability (i.e., it includes considerations of economic growth, well-being, and ecosystems), the nuances lie in whether the authors are focused on a) proposing a new discipline or sub-discipline, or b) reviewing experiences of the interaction between social sciences and marine studies.

Marine Social Sciences: Discipline or Sub-discipline?

It is useful to have a notion of the social sciences as understood by the academic circle cited here. In the most general sense, Flyvbjerg (cited in Bavinck et al., 2020) describes them as practical wisdom for addressing and acting on social problems within a given context. Regarding their areas, Smith (2002) classifies them into two groups: the first includes psychology, anthropology, sociology, political science, and law, collectively encompassing humanity, the body, society, and decision-making. The second group includes economics, geography, history, and archaeology, where there is an emphasis on the material relationships of humanity in space and time. In any case, they are

[4] This, according to the understanding of modern societies outlined by Habermas in his Theory of Communicative Action. Culture is articulated into validity aspects around issues of truth (science and technology), justice (law and morality), and taste (art). The basic institutions of society (family, church, legal order, etc.) form functional systems (economy, state administration). Finally, the individual undergoes processes of socialization that endow them with personality structures and serve for autonomous orientation in their adult life.

granted privileged access to understanding the complex interplay between social dynamics and environmental contexts human behavior for specific purposes —in this case, to analyze and change the relationship with the marine environment.

The diagnosis is that this relationship has become problematic not only due to undesirable consequences such as pollution and conflicts of interest but also from the very conceptualization. Proponents of creating a new discipline or deriving a sub-discipline (primarily from sociology or ethnography) criticize the land/sea dichotomy[5] because the complexity of the "global ocean" has emergent processes that transcend a sharp differentiation: Should the paradigm of "terrestrial sociology" be extended to the marine realm? Should such phenomena be explained from a terrestrial perspective? Is a uniquely marine theory possible? (Cocco, 2013; Kolodziej, 2020 & 2022). Additionally, maritime topics have historically been dominated by the natural sciences, leaving human dimensions relegated (McKinley et al., 2020; Van Putten et al., 2021).

Historical perspectives provide insights into the attempts to propose a new discipline or derive a sub-discipline. The earliest effort comes from William R. Rosengren, who, in a 1973 editorial titled "Sociological Approach to Maritime Studies: A Statement and Example," largely outlined the general agenda as understood today. He recognized a plurality of interests attempting to establish definitions of the marine environment and its specific uses. He also emphasized the importance of analyzing non-economic rationalities related to the sea, such as symbolic values and aesthetic implications at local levels.

It wasn't until 1981 that Michael Poole, in "Maritime Sociology: Towards a Delimitation of Themes and Analytical Frameworks," revisited the definition and delimitation of maritime sociology. He shared with Rosengren a sociological theoretical framework: "the social and cultural forces that determine definitions of the marine environment, which in turn shape its use and exploitation" (p. 2). Poole provided a list of topics for analyzing this environment: the concept of the social system, community and association, social action and culture, differentiation and stratification, processes, change and technology, and conflict and deviance. Thus, in the internal historiography of attempts to establish a new discipline or sub-discipline, Rosengren and Poole laid foundational work that contemporary scholars continue to build upon. In the historiography of maritime social sciences,

[5] Picken points out that "Underwater realities mobilize the binaries and dualisms that make sense in social space, highlighting how these are historically situated on land" (2014, p. 12).

Poole and Rosengren emerge as precursors of the discussion (Cocco, 2013 & 2015; Kolodziej et al., 2015 & 2022; Hannigan, 2017).

However, there is a parallel history in Poland. There, interest in maritime sociology has been present since the institutionalization of sociology after World War II, with Ludwik Janiszewski as its main proponent. From a theoretical standpoint, he introduced the concept of "marinization" in 1971, describing the process through which human activity relates to the sea (socio-culturally), influences social reality, and fosters the emergence of marine elements in both material and non-material aspects (Kolodziej, 2020). Although his approach emphasizes the impact of general economic activity, it aims to name an ancient social process (predating industrialization and urbanization) that permeates humanity. Among the approaches to the subject, this one shares more characteristics with a general theory of society rather than an analysis of social subsystems related to marine activities.

John Hannigan (2017) acknowledges that while maritime sociology has addressed social realities surrounding the ocean – such as maritime professions, fishing and seafaring families, port cities, and naval culture – since at least the mid-20th century, new contexts – ecology, piracy, immigration, economy, and management – warrant formulating a critical political economy to address the current state of capitalism and the ocean crisis along with its accompanying geopolitics. In the realm of human-nature relations, he considers that post-humanist currents aptly critique the economic order. Nonetheless, Hannigan clarifies that his proposal (Sociology of the Oceans) serves more as a thematic glue than a conceptual or theoretical body. In other words, it would be a sub-discipline of environmental sociology that, nonetheless, would address the qualitatively different properties of the ocean as a space being constructed and contested.

The perspective that the analysis of capitalism deserves central attention is shared by several authors (Clark, 2008; Longo, 2016; Bavinck, 2018; Lambach, 2022). In this vein, and with respect to Latin America, the works of Gustavo Moura from Brazil and Nemer Narchi from Mexico propose a human or social oceanography. Both, from the perspectives of human geography, ethnography, and anthropology, emphasize the colonial and Eurocentric legacy of oceanography, the technical-instrumental outcome of its inquiries, and how these, in service of industrialization, have caused environmental degradation and a history of dispossession of local coastal communities. To counter this, it is a priority to enhance marine databases with local knowledge; this, in turn, supports ethno-sciences which have epistemological (structuring the production of knowledge), philosophical (how

reality is perceived), and cognitive (discovering universal patterns of ordering reality) value (Narchi, 2010). Moura (2020), from the perspective of socio-environmentalism, critiques the reductionist nature of modern rationality and advocates for an "indigenous peoples' political praxis" that, in its local knowledge, inherently includes a historical protection of biodiversity.

The Ocean as a Social Space

A common theme in the reviewed literature is the conceptualization around "the construction of a social space." From our own perspective, this can be explained by various factors: the influence of the linguistic turn in philosophy since the early 20th century (Habermas, 2002), the particular success of Peter L. Berger and Thomas Luckmann's book “The Social Construction of Reality” (1966), and a specific context: "[social construction] is one of the many ideas being fiercely contested in the American culture wars" (Hacking, 1999, p. 11).

The concept of social space "makes reference to the sensually perceptible, imaginations, and memories as a product of society producing its own specific space. Löw develops an understanding of space as relational arrangements of social goods and living beings at places" (Kolodziej, 2022, p. 17). Based on this framework, we find books such as “Framing the Ocean, 1700 to the Present: Envisaging the Sea as Social Space” (Gordinier, 2015) and “The Social Construction of the Ocean” (Steinberg, 2001). Emilio Cocco argues that "in a truly maritime sociological approach, the sea is not only a medium but a social space, which would not be merely ‘used by society’ but rather represent ‘a space of society’" (2013, p. 5). Crockford, highlighting what kind of ocean has been emphasized in its construction, underscores an extractivist root: the foundation of the "Blue Economy" lies in the techno-science of deep ocean engineering, and the social license to operate, developed through corporate social responsibility, is merely an extension of terrestrial mining practices (2020).

This perspective sharpens the critique of conceptions of the global ocean that serve industrialization, military uses, or spatial appropriation. Lambach (2022) helps us distinguish between two functions of technology in the construction of oceanic space: a causal function that produces new spaces for specific uses, and an interpretive function that imbues these spaces with social meanings and values or facilitator, and another as a medium. In the former, technology creates potentials for action by modifying a scale (e.g., improved bathymetry enables the identification of extractive resources). In the latter, the various modes of ocean representation act

as a bridge between interested parties and the environment: coastal mapping can have different strategic uses depending on whether the interested party is an NGO or an oil company.

When specifying the tools used in theorizing the social construction of oceanic space, we find borrowings from systems theory, economic geography studies on the spatiality of capitalism, political geography on territoriality, and poststructuralist concepts on social conflict in the production of space. All of these aim to renew a critique of territorial economic policy (Steinberg, 2001). Regarding the social sciences, Steinberg notes, "Within the social sciences more generally, the bulk of research has focused on one or another use of the marine environment, but not on the ocean as an integrated space that is a product of – as well as a resource for – a variety of human uses" (p. 10).

From the perspective of its proponents, viewing the ocean as a social space aids the normative frameworks of ecosystem conservation in analyzing conflicting actors and interests to reach consensually approved decisions. Thus, "New marine spatial planning (MSP) policies seek to categorize and represent ocean spaces and activities in an effort to provide a solution to long-standing controversies stemming from individual sector-based management" (Boucquey et al., 2016). This approach aims to identify where biases favoring one appropriation over another exist within the conceptualization and practices of maritime management. According to Boucquey, the same tool can serve both to reinforce an exploitative attitude toward the environment (neoliberalization of nature) and to expand the capacities of a community and other environmental actors.

Past Experiences: Fisheries and Social Sciences

If there is any area within the maritime universe where the interaction with social sciences can be documented, it is in fisheries. Since the 1970s, the research agenda in this field has expanded due to formulations on the management of common resources (Elinor Ostrom) and the co-management theory (public administration and fisheries) of Jentoft (Symes, 2010). One of the landmark works of that era is James M. Acheson's called “Anthropology of Fishing” (1981). For some, it is clear that decades of experience in the social sciences should suffice to address the challenges posed by current and future marine industrialization: "... marine social science must enhance understanding

of the dynamics of industrial development in ocean and coastal spaces, including their governance contexts" (Knol-Kauffman et al., 2020).

Another guiding element within this ocean of perspectives is the recognition of two main trends: conservation and industrialization. Arbo et al. (2018) outline that interests around the sea are concentrated on "Blue Growth" and environmental concerns. In this arena of contention, natural scientists, often with substantial help from social scientists, tend to interpret the ocean as a sick entity and a valuable natural and cultural heritage that must be protected and restored. Conversely, industrialists, particularly engineers, emphasize the sea as an economic frontier with much untapped resource potential. It is important to remember that these typologies serve to simplify the discussion, as in reality, there is a blend of perspectives.

Amidst these conflicting interests, social sciences are projected as a pillar in management based on scientific recommendations. One interpretation is that social sciences are essential for understanding and intervening in modernization practices and limiting the privatization of resources. Consequently, they are seen as having the potential to formulate governance alternatives, anticipate future trends, envision desirable futures, and facilitate just social processes (Arbo et al., 2018). This role can be categorized as strengthening capacity-building in marine and coastal management. Smith (2002) comments, "Perhaps the best —and most problematical— example is fisheries management, where large-scale biological research programs input into decision-making which also involves technical gear components, the economics and management of the industry itself on both catching and processing and marketing sides, and political and legal inputs as well."

But what has become of the social science programs actually implemented in fisheries management and orientation? As noted in footnote 2 of this article, the presence of social science departments within the UN structure throughout history might surprise those who perceive social sciences as being sidelined in marine matters. Significant examples include the establishment of the European Social Science Fisheries Network in 1996, aimed at socially orienting fisheries research (community-based solutions, local co-management, and rights-based approaches, territorial usage rights, community quotas, etc.) (Symes et al., 2010). This effort was significant, leading to the establishment of the MARE (Maritime Research) Center and symposium in 2000, which brings together social scientists, organizes courses and international conferences, and publishes a journal with a broad focus on social sciences, not just fisheries, called “Maritime Studies”.

Arbo (2018) even asserts that these decades of interaction between social sciences and fisheries have developed a critical niche for understanding transformations in the ocean and coastal economies, housing, cultures, and social practices, although he notes this is within the context of Northern Europe. This review is supported by other evaluations of the subject. For instance, the article "A Decade Incorporating Social Sciences in the Integrated Marine Biosphere Research Project (IMBeR): Much Done, Much to Do?" (Van Putten et al., 2021) highlights ten years during which social scientists have been involved in regionally based science programs[6]. Within this conglomerate, there was a section called the Human Dimensions Working Group, whose task was to understand the interactions between human and oceanic systems, recognizing that humans not only influence the marine system but also depend on it for goods and services.

The perceptions among participants in these programs suggest that for greater connection between scientists of different disciplines, there needs to be established research priorities, access to funding, interdisciplinary communication, and a general awakening of interest. In this regard, several authors have focused on generating an agenda of research topics and priorities (McKinley et al., 2020 & 2022; Bennett, 2019; Fish, 2011; Smith, 2002; Bavinck, 2018; Silver, 2015). Another revealing perception, particularly regarding real interaction between natural scientists, engineers, and social scientists, is that the former, while recognizing the value of sociology, anthropology, history, and other disciplines, think that these may not be relevant to all the objectives and activities of regional programs (Van Putten et al., 2021, p. 9). They perhaps see more clearly the relevance of social knowledge in legal matters, public policy, and science communication.

According to another comprehensive survey (1200 participants) aimed at defining important issues in maritime matters (Lacroix et al., 2016), nine main challenges were identified (p. 6):

1) Understanding the evolution of marine ecosystems

2) Ensuring the supply of food and therapeutic products from the sea

3) Improving ocean colonization

4) Developing technologies for the exploitation of marine resources

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[6] The historical sequence of the Integrated Marine Biosphere Research Project is as follows: in 2002, the International Geosphere-Biosphere Programme (IGBP) and the Scientific Committee on Oceanic Research (SCOR) created the Integrated Marine Biogeochemistry and Ecosystem Research. In 2016, the name was changed to Integrated Marine Biosphere Research, reflecting a focus on the human dimensions of global change and ocean sustainability. It continues to be funded by SCOR and is part of the Future Earth Global Research Projects.

5) Risk prevention

6) Developing biotechnologies for industrial and health uses

7) Developing education, sustainable tourism, and responsible social practices

8) Establishing international maritime law adapted to current circumstances

9) Systematizing and globalizing the monitoring and control of marine activities

The conclusion is clear: the maritime powers —referring to the United States, the European Union, and China in this study— show a clear interest in securing their national resources rather than promoting international cooperation for safe trade and sustainable exploitation.

Reflection from the fisheries sector reappears in this context. Even within governance frameworks that include social sciences, there are indications that social aspects and the dimension of knowledge are inadequately theorized and represented (Jentoft et al., 2014). Improvements in governance schemes include contributions regarding the conceptualization of "community," as there are often problems of exclusion due to generalization in fisheries. Communities and their relationships are highly diverse, so a typology encompassing modes of life, livelihood, or work could better inform environmental policies[7] (Delgado et al., 2022). This concern, with an emphasis on communities, is present in social impact analyses (due to undesirable dependency on "green infrastructures") (Arkema et al., 2017), their integration into a "trans-global oceanic world" (Wozniak, 2013), and attention to "missing layers" of analysis.

Marine Social Sciences: A Panoramic Reflection

The current enthusiasm for reflecting on social sciences and maritime studies appears to stem from various sources, as observed in this review: from the proposals for a new discipline or sub-discipline, the influence of social construction theory of spaces, criticism of capitalism, and fisheries-focused approaches. Undoubtedly, all these efforts aim to foster a change in human relationships with the

[7] In fact, this kind of criticism was the basis for the 2002 reform of the political-legal framework of the European Union known as the Common Fisheries Programme (CFP), as it brought about problems such as a highly centralized decision-making system applied extensively to highly diverse European fishing areas, both politically and ecologically. The reform introduced an ecosystem approach with emphasis on resource conservation and the development of long-term management plans, as well as the establishment of networks of Regional Advisory Councils that brought together key local stakeholders in specific areas to provide practical advice on fisheries public policies (Symes et al., 2010).

marine environment, whether through conceptualization or the implementation of governance tools. Viewed from a systemic perspective, the increasing complexity due to the multitude of competing interests and the vast empirical and physical knowledge of the ocean makes sustainability goals a Herculean task. Social sciences are called upon primarily for their capacity to decipher issues related to behavior, institutional development, and impact assessments (Symes et al., 2010).

Given the highly diversified nature of this field, the very plurality of perspectives in social sciences is a controversial element in itself. Advocates for creating a new discipline or sub-discipline are not only interested in defending the specificity and relevance of social analysis regarding oceanic basin issues but also see the opportunity to renew social sciences through the metaphors and qualities evoked by the ever-moving water: fluidity, plasticity, resistance to control, etc. (Cocco, 2013; SSaSea, n.d.; Picken, n.d.). Other perspectives, even within the governance framework and its practical demands, express dissatisfaction with reducing social sciences to mere development indicators or impact assessment tools (Moura, 2020; Arbo et al., 2018).

In the quest for a foundation of social knowledge, there are proposals that attempt to include philosophical perspectives on the issue. See, for example, the text "Five Questions to Understand Epistemology and Its Influence on Integrative Marine Research" by Katie Moon and others (2021), or the inclusion of the term "phronetic dimension" (referring to Aristotle's philosophy) in Jentoft's (2014) reflection on knowledge in fisheries governance in the European Union. There seems to be a struggle and difficulty in proposing new approaches that simultaneously contain higher-order organizing elements. This challenge is not only theoretical but also evident in the names given to various attempts to create communities of social researchers interested in the sea.

John Hannigan (2017) has noted that several other proposals have failed to take off, and his rebranding of Maritime Sociology as the Sociology of the Oceans seeks to avoid the same fate. Nonetheless, groups have emerged and disappeared without significant impact. Even those with institutional support and specific funding —such as the European Social Science Fisheries Network, previously mentioned— have faced endings that leave questions. Despite three years of European Commission funding, which allowed it to hold several workshops on socially-oriented policy, gather a core group of 30 researchers, and develop an extensive publication program, it failed to provide a lasting institutional framework (Symes et al., 2010, p. 270).

Some of the currently active academic communities focused on the interaction between social sciences and maritime studies include the Marine Social Science Network, the CoLaboratory for Social Oceanography, the Maritime Sociology group at the Institute of Sociology at the University of Szczecin (Poland), and the Centre for Maritime Research (MARE). These groups maintain their identity as active networks with mailing lists, regular communications[8], conference calls, publications, and a more or less defined focus.

Regarding the breadth of this bibliographic review, even having considered academic groups from Latin America and Poland (omitted in McKinley's survey in the article "Development and Expansion in the Marine Social Sciences: Insights from the Global Community" [2022]), a more in-depth accounting of interest groups in the Asia-Pacific region is still needed. We know that in China, since 2001, the Institute of Marine Culture and Society was founded (Kolodziej, 2022), and the Chinese Sociological Association has already established professional committees in maritime sociology (Hannigan, 2017). The same diagnosis seems to hold despite the "cultural distance" from the West: "a sociological perspective is needed to inform management, with social scientists being required to balance the over-confident other scientists in other fields because the importance of knowledge and the role of 'experts' is deeply social" (Ning'er cited in Hannigan, 2017).

Paradoxically, social sciences seem to be experiencing a new boom due to the demands arising from oceanic issues, but simultaneously, they face a demand for accreditation: their value, utility, and practical application. This is part of their disciplinary history. Categories and proposals for new approaches continue to proliferate, whether from radical critiques of the system or frameworks adapted to capacity-building in specific governance tasks. In any case, the struggle to consolidate the ocean as a socio-environmental object is underway, presenting an opportunity to test interdisciplinarity within the grand context of human history with the sea.

[8] The journals that accept perspectives on social sciences and marine topics are numerous. Some examples include Maritime Policy & Management, Maritime Studies, Transnav: International Journal on Maritime Navigation & Safety of Sea Transportation, Constanta Maritime University Annals, Pomorstvo/Scientific Journal of Maritime Research, Scientific Journals of the Maritime University of Szczecin, Journal of Coastal Research, Asia-Pacific Journal of Marine Science & Education, and International Maritime Health.

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